Authors' Instructions: Preparation of Camera-Ready Contributions to SCITEPRESS Proceedings

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Keywords: The paper must have at least one keyword. The text must be set to 9-point font size and without the use of bold or italic font style. For more than one keyword, please use a comma as a separator. Keywords must be titlecased.

Abstract: The abstract should summarize the contents of the paper and should contain at least 70 and at most 200 words. The text must be set to 9-point font size.

1 Introduction

>>> To be completed...

- We present "search coaching" as a way to instill knowledge faithfully to a learner;
- We compare this to our own and others' previous works;
- We maybe discuss a bit about LLMs. (?)

2 Related literature

>>> To be completed...

- Related works as in the proxy coaching paper;
- A brief update on what is the current state of the art;
- Maybe some works that draw links to the planning aspect of this paper.

3 Experimental setup

In this Section we empirically explore search coaching, by having a machine learner iteratively seek advice by a coach, as in (**?**), with the aim of iteratively improving its search capacity, as per the coach's explicit search policy. Given a fixed goal state, g, the learner applies its search policy, $\mathtt{P}_{learner}$, to identify a path, p, starting from an arbitrary start state s and ending at g. Upon lack of knowledge, the learner seeks advice from its coach, which responds according to its own policy, $\mathtt{P}_{coach}$. A *coaching session* corresponds to a full learning cycle, where the learner eventually crafts one such path or the process is aborted.

Add a few words about ho
in the context of searching
past sub–optimal state the
piece of advice according
explicit) policy that improve
a(n implicit / explicit) goal.

3.1 Sorting coaching

In this series of coaching sessions we have implemented two variants of a machine coach, aiming to explain to an initial ignorant agent about how to sort a list of numbers, as a rough equivalent to linear preference elicitation. The first coach, $\mathcal{C}_{\text{bubble}}$, uses bubble sort as its underlying policy while the second one, $\mathcal{C}_{\text{quick}}$, uses quicksort to provide advice from. We explored different state sizes, n, ranging from 1 to 20, running $m = 100$ iterations for each value of n. We have also explored two different types of advice for each coach: (A1) *full advice*, where each piece of advice depends on the entire state, and; (A2) *partial advice*, where each piece of advice depends on the specific part of the state to be improved. Moreover, for each advice type we have considered three learner configurations regarding advice memory: (M1) *no memory* across different iterations for the same state size, n, i.e., coaching sessions are pairwise independent; (M2) *short memory* across different iterations for the same state size, n, i.e., coaching sessions

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across different values of n are independent but not within the same value for n, and; (M3) *long memory* across all values of state size, n, so all coaching sessions are correlated, building on knowledge from previous test cases. In all cases the learner seeks advice upon lack of knowledge for further searching, returning a partial path p starting from s, while the coach adopts one of the following *coaching styles*: (C1) in *reactive* coaching, denoted by a superscript a, the coach provides advice on learner's last wrong decision in p; (C2) in *proactive* coaching, denoted by a superscript p, the coach provides advice on learner's first wrong decision in p; (C3) in *reflexive* coaching, denoted by a superscript r, the coach provides advice at a random wrong decision in p.

In Figures 1 we present the number of coaching interactions required to reach the goal state in each case as a function of the state size, n. Regarding the full advice condition (Fig. 1a), the utilization of either short of long term memory does not appear to have any particular effect on the number of coaching interactions, which was expected given the overspecific nature of provided advice, being fully tailored to the state it triggered the learner to seek for that piece of advice. On the contrary, partial advice (Fig. 1b) offer a significant advantage to the learner, in both short and long memory conditions. As for the short memory case,

Data: s, g, $learner$, $coach$
Result: $learner$ with updated searching policy.
$advice \leftarrow$ `null`;
do
  $learner.update\_hypothesis(advice)$;
  $path \leftarrow learner.search\_path(s, g)$;
  $advice \leftarrow coach.evaluate(path, s, g)$;
while *advice is not* `null`;

Algorithm 1: Coaching process. s, g denote the start and goal states respectively.

3.2 Clustering Coaching

- We explore clustering in various settings, using an inertia-based algorithm.
- The setting is **similar** in terms of the coaching process, but there do exist **domain differences:**
  - Algorithms rely mostly on full– and not partial–state advice;
  - We are exploring how different levels of "generalization freedom" affect learning (with memory).

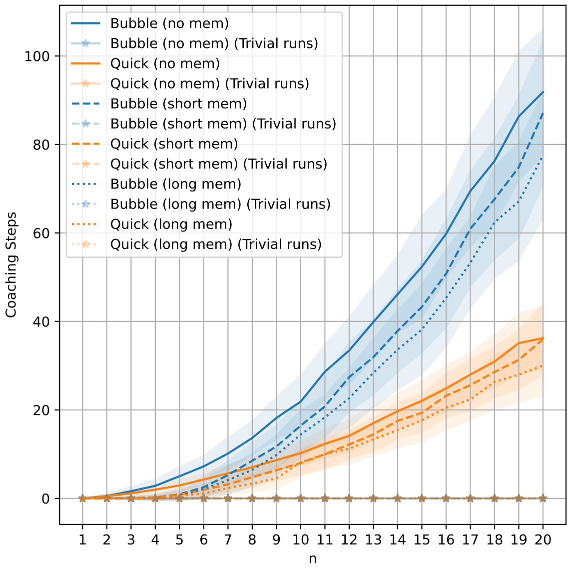
4 Results

>>> To be completed...

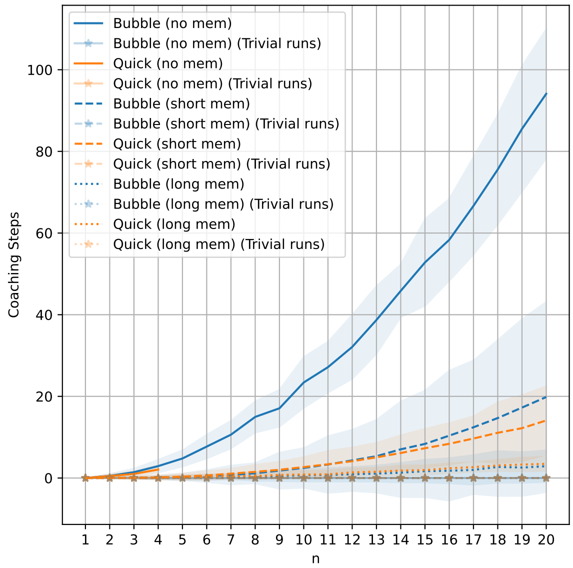
- Find a metric to quantify learner's fidelity (brainstorming follows):
  - Maybe the number of advice itself is an indicator, at least in terms of efficacy – it still might be the case that the learner achieves the same result through another path;
  - Different coaching styles (reactive, reflexive, proactive) should affect faithfulness / fidelity, so this metric should capture it.
  - The absolute (or relative?) frequency of differences between the learner's and the coach's paths on a given benchmark set could be the most fine grained metric. So, for instance, if the learner goes $s \rightarrow a \rightarrow b \rightarrow c \rightarrow g$ while the coach would go $s \rightarrow a \rightarrow b \rightarrow d \rightarrow g$, we have an absolute (relative) difference of 1 (0.33, counting only intermediate nodes).
  - In the above, we might also count the goal node in the relative frequency case. Also, we should somehow measure distances among paths of different length.

5 Conclusions and future work

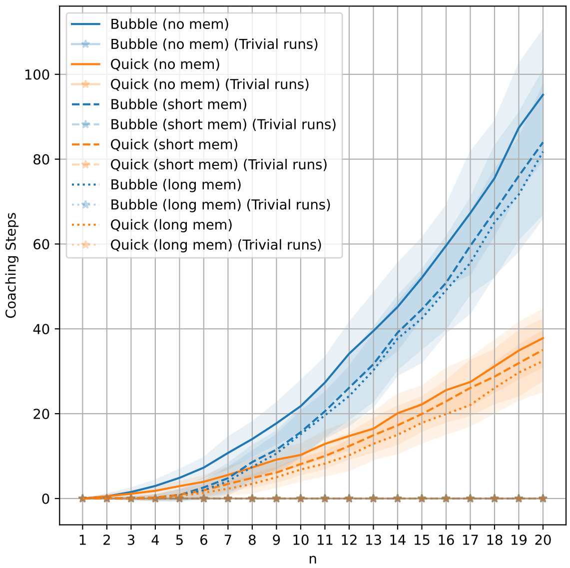
>>> To be completed...



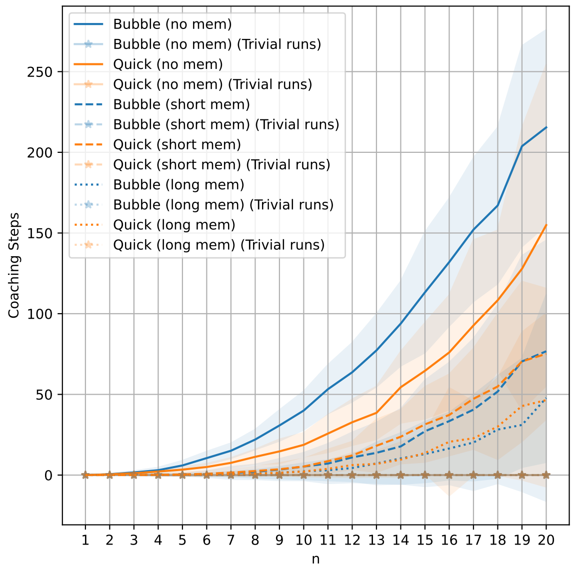
(a) Full condition advice, reactive coaching



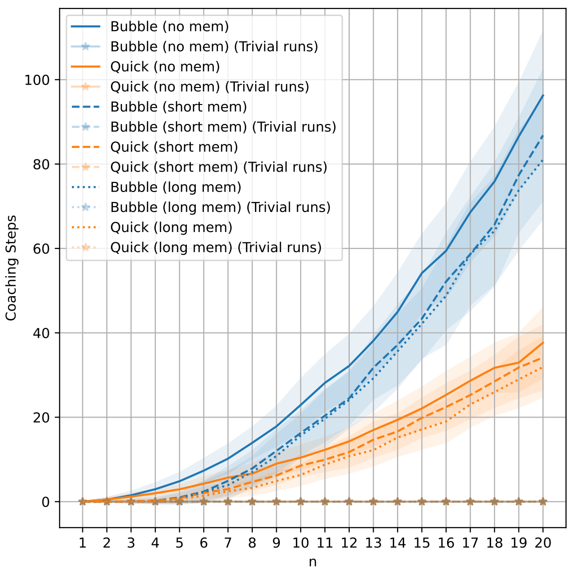
(b) Partial condition advice, reactive coaching



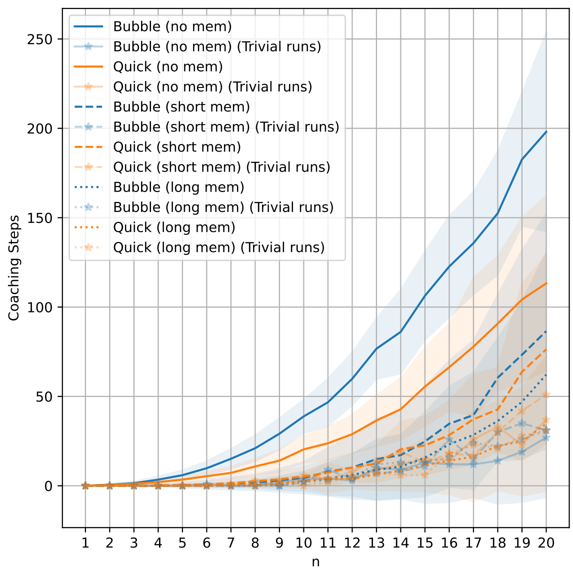
(c) Full condition advice, proactive coaching



(d) Partial condition advice, proactive coaching



(e) Full condition advice, reflexive coaching



(f) Partial condition advice, reflexive coaching

Figure 1: >>> To be completed. . .